Evolution of Cosmic Structure

Lecture 12 - Feedback in galaxy formation

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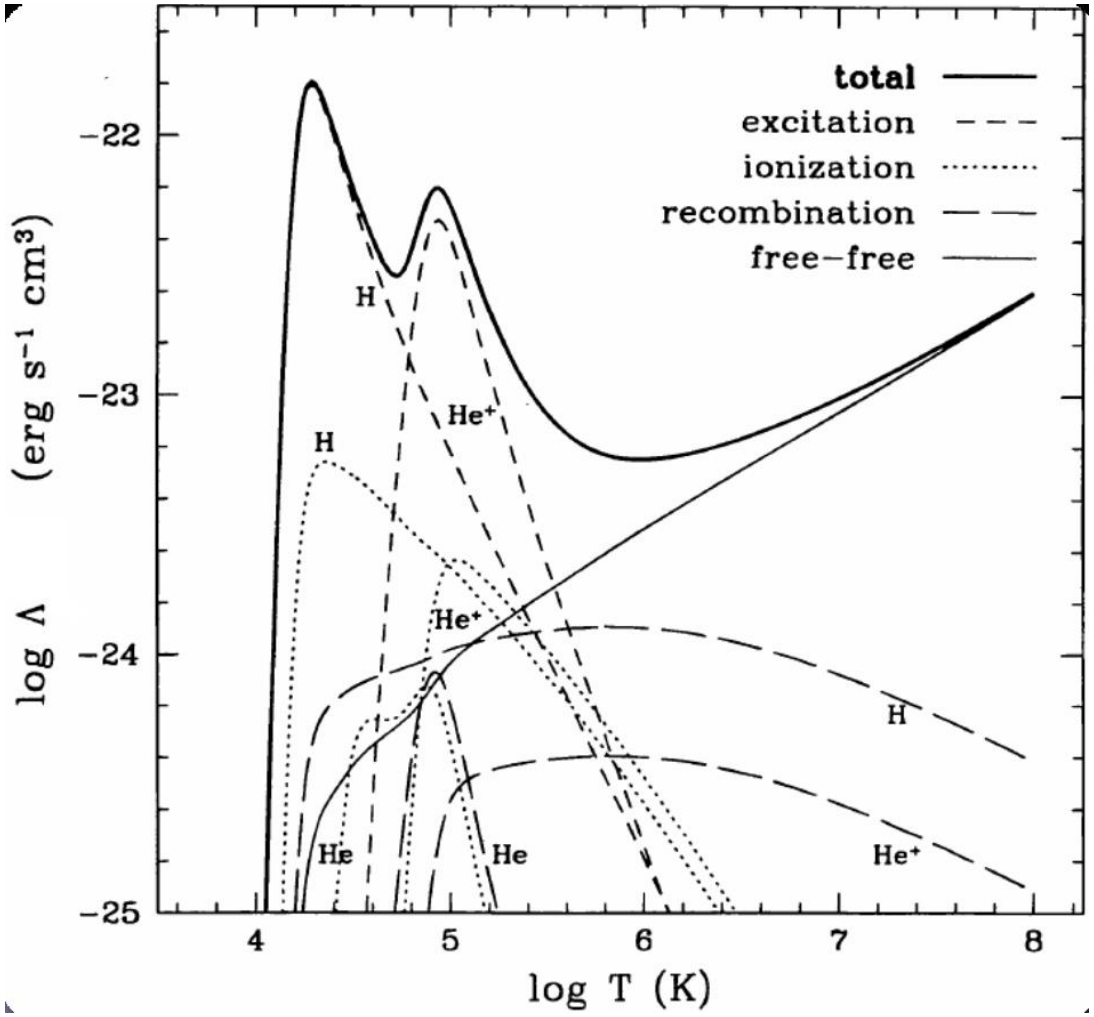
Learning Outcomes

What techniques are used to simulate the development of cosmic structure? How can complex baryon physics be incorporated? What light do the results cast on the evolution of structure?

- The role of simulations
- N-body simulations and dark matter halos
- Adding gas – hydrodynamics
- Incorporating baryon physics
- The semi-analytic approach
- A physical model of galaxy formation – cooling, feedback

Cooling equilibrium and cooling function

The cooling rates depend on the detailed ionic species for each of the atomic species. For instance, n_e, n_{H0}, n_{H+} for hydrogen. The creation and destruction of those species is short compared to the hydrodynamics, so ionization equilibrium is usually assumed (either photoionization or collisional).



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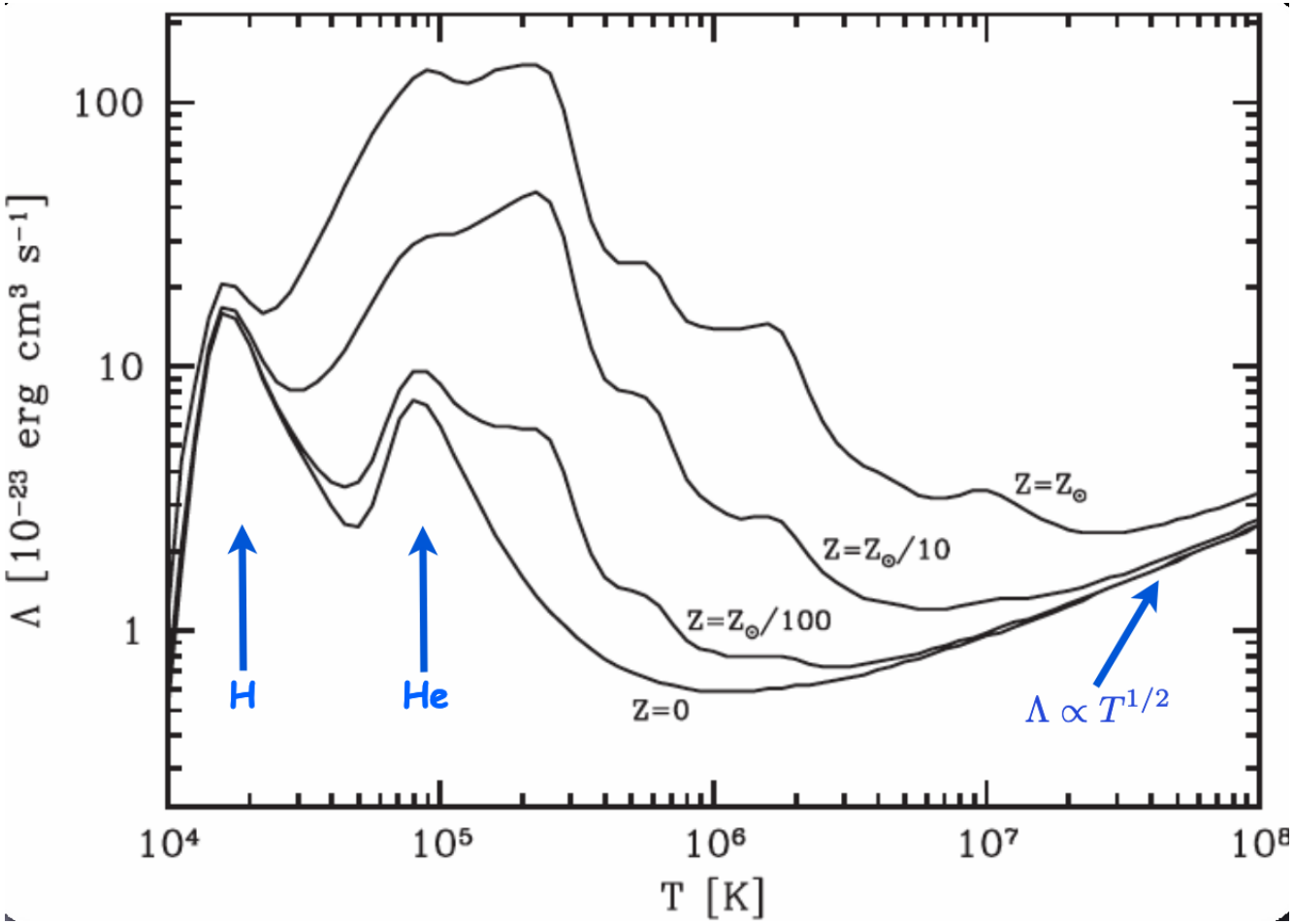
At high T, gas is fully ionized, - only bremmstrahlung

At T < 10^4 K, all gas is neutral – no cooling

At t ~ 10^4 K, all H is ionizied – cooling peak

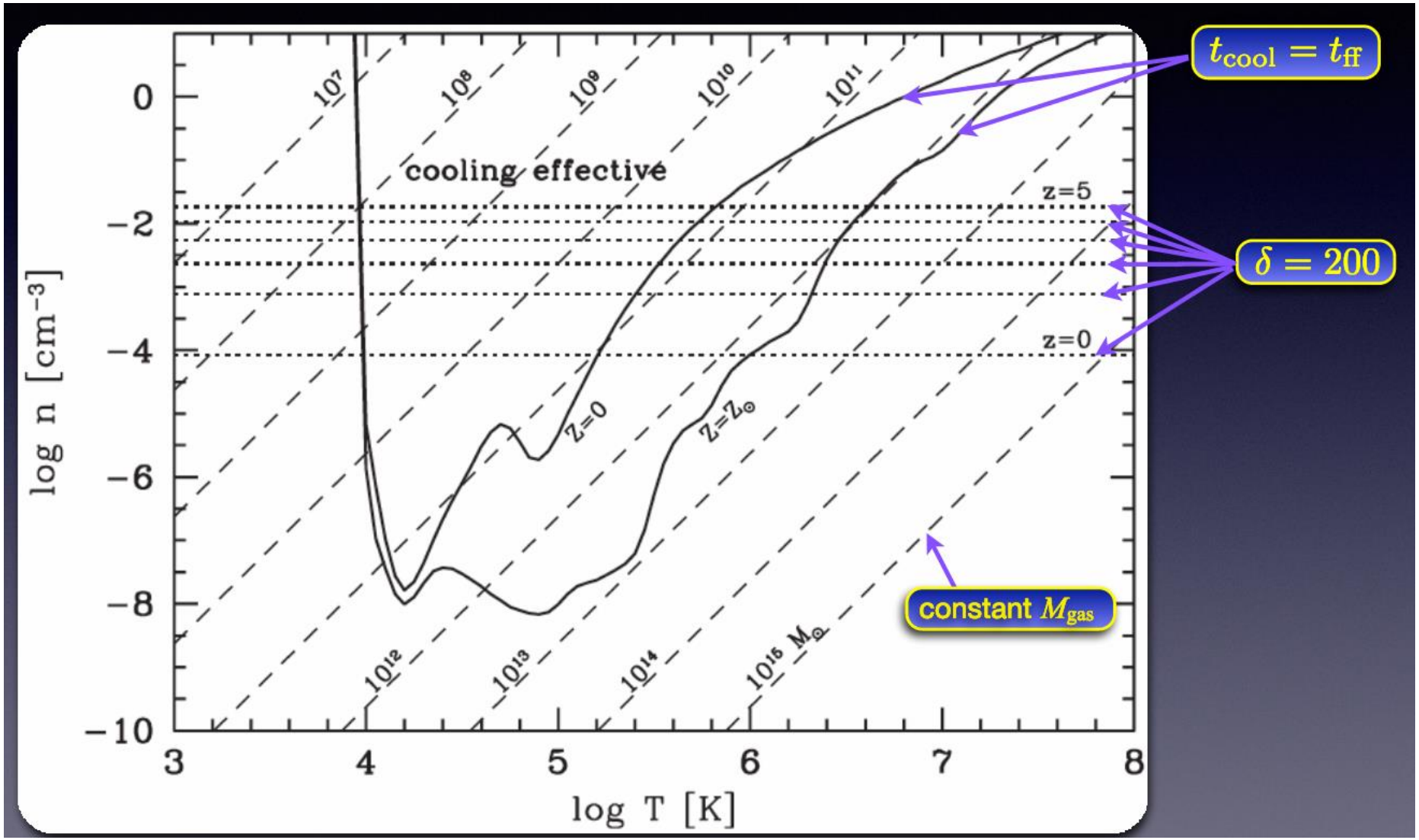
At t ~ 10^5 K, He is ionized – second peak.

Other metals contribute depending on their structure and abundance



Sites of galaxy formation

We can plot the location of the Tcool = tff lines in density - temperature space to see the locations of galaxy formation



Neither small or large mass halos can efficiently form

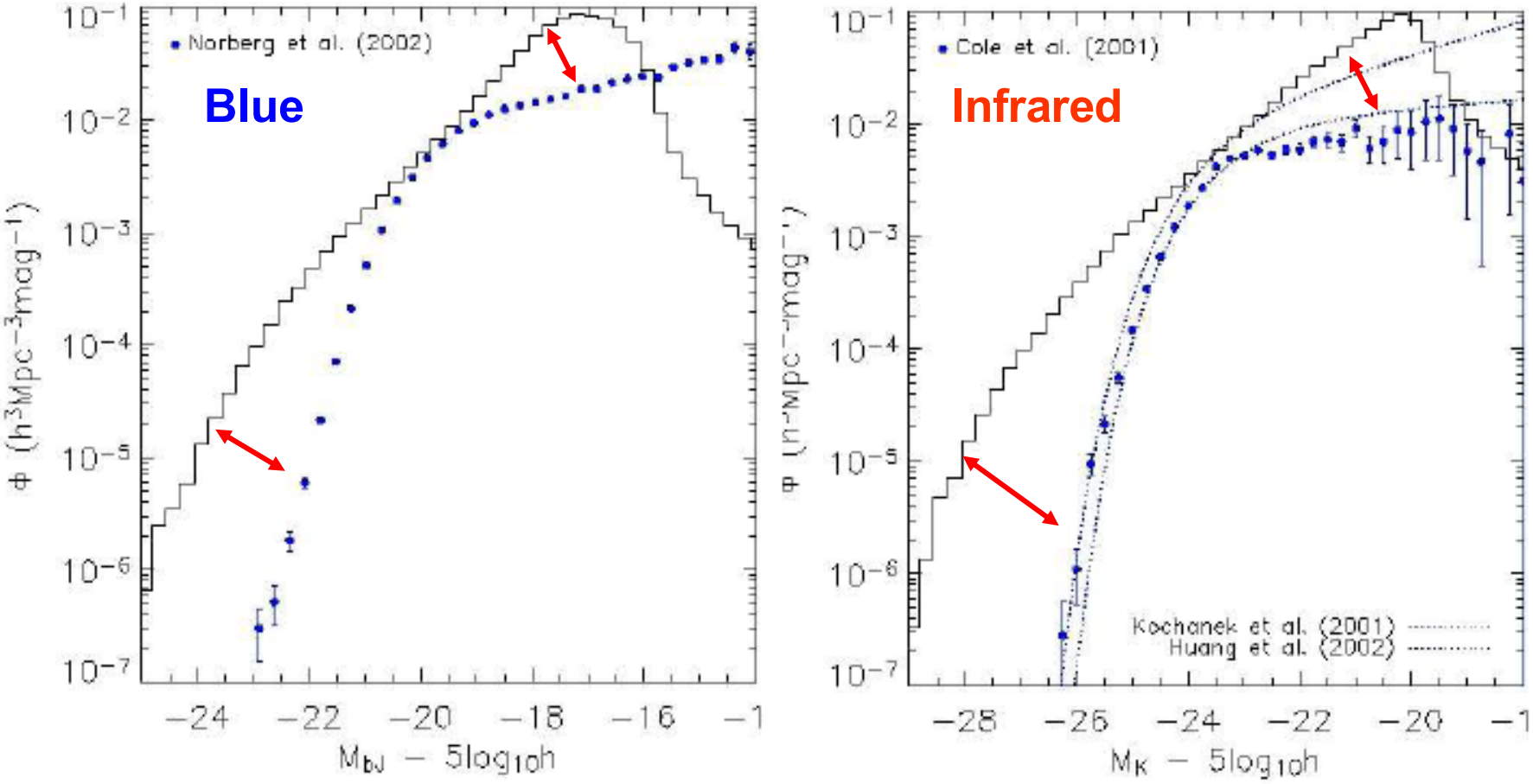
The galaxy luminosity function

This predictions for a galaxy luminosity function has been extensively investigated using semi-analytic models - see e.g. Benson et al (2003), Croton et al (2006), De Lucia et al (2006), Bower et al (2006) – and in hydrodynamical simulations e.g, Schaye et al. (2015), Vogelsberger et al. (2014) -- and it seems likely that *cosmic feedback* is affecting the baryons within dark halos. In particular

- Heating of gas during reionisation may prevent gas cooling and forming stars within small halos
- Supernova feedback can blow gas out or heat gas in fairly small galaxies, limiting the efficiency of star formation
- Feedback from AGN limits the ability of gas to cool and form stars at the centres of group and cluster mass halos

Some examples from the model of Croton et al follow.

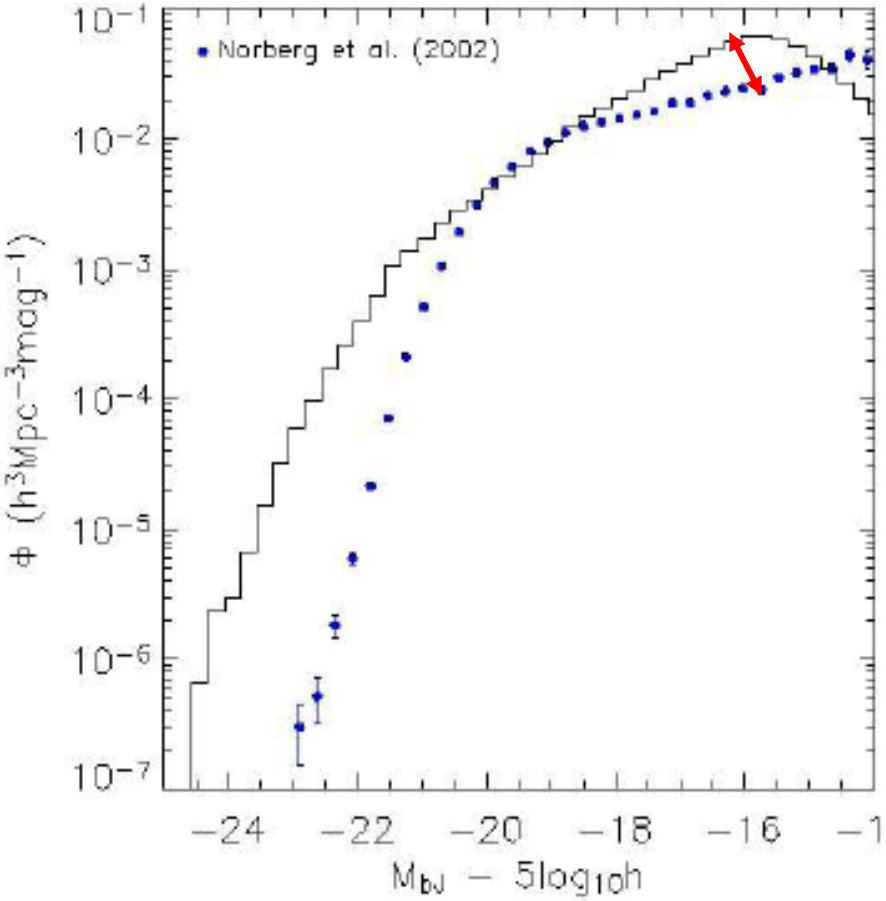
Effect of feedback on the Luminosity Function

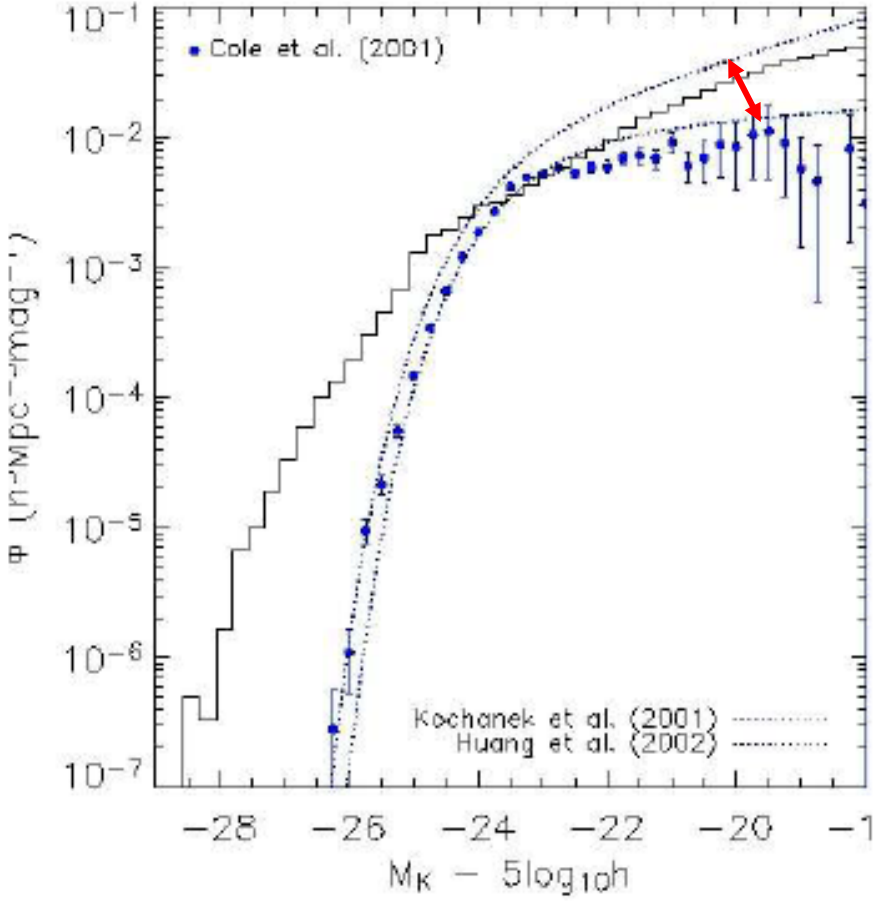


Model with no reionisation input or feedback.

Croton et al 2004

Effect of feedback on the Luminosity Function

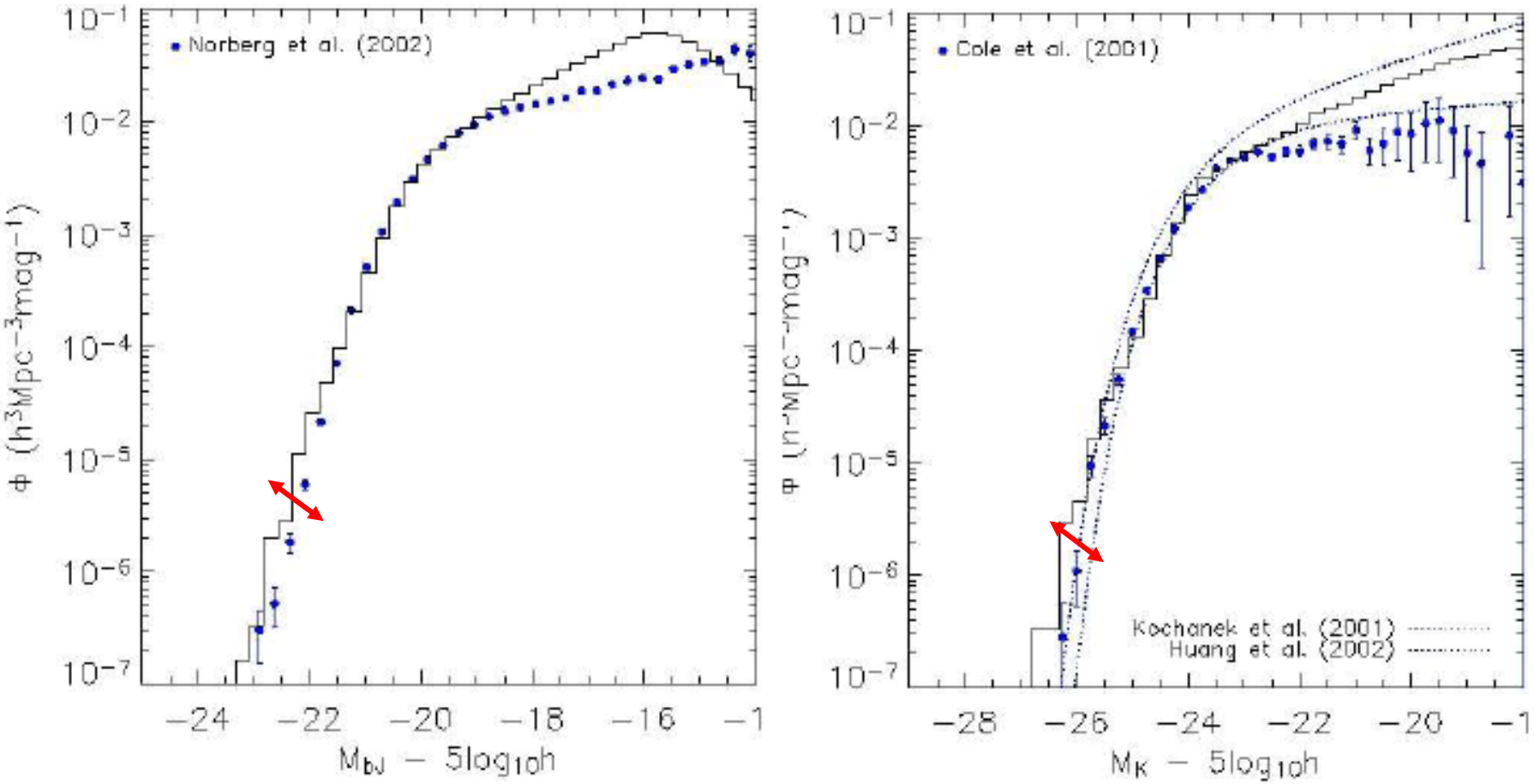




Model with SN feedback.

Croton et al 2004

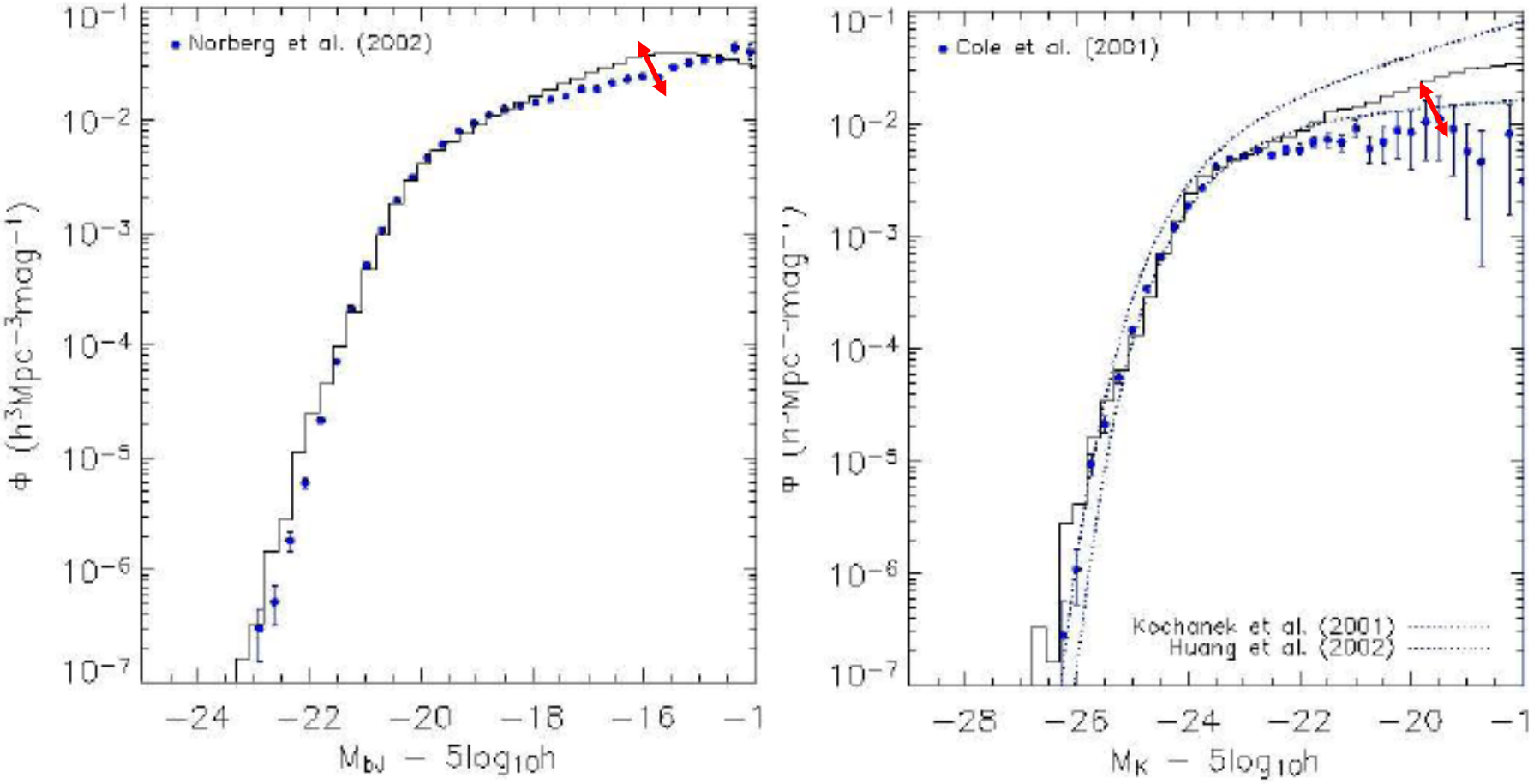
Effect of feedback on the Luminosity Function



Model with SN and AGN feedback.

Croton et al 2004

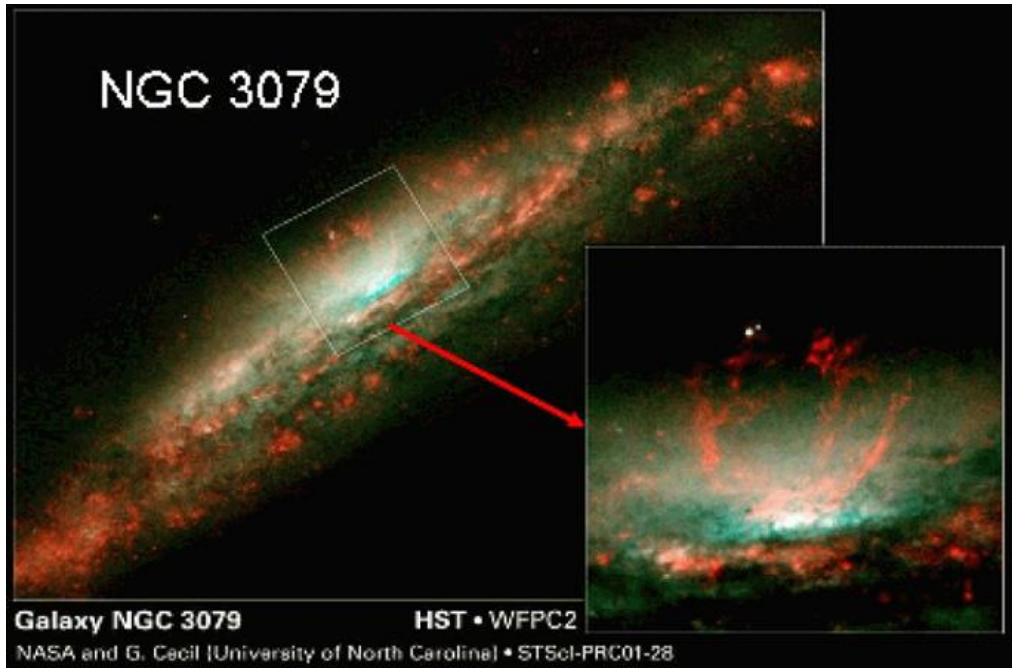
Effect of feedback on the Luminosity Function

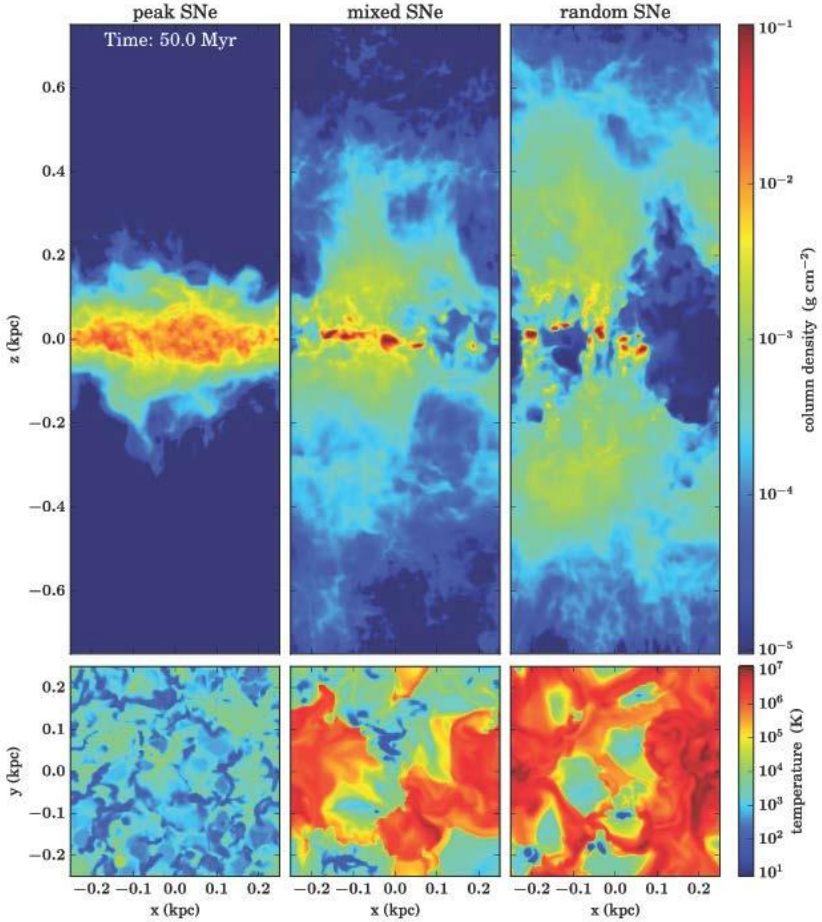


Model with SN and AGN feedback and reionisation heating. Croton et al 2004

Observations of Supernova feedback

Direct observations are difficult, but there are outflows which suggest a SN origin, and triggered explosions in simulations can mimic that. Supernova feedback could be through the coupling of the gas ‘kinetically’ or ‘thermally’.





Kinetic supernova feedback

Kinetic feedback – gas is ejected out of the dark matter halo and can not form future stars.

What is the escape velocity?

Requires energy

$$E_{ej} = \frac{1}{2} m_{ej} V_{esc}^2$$

Escape velocity of an NFW halo is

$$V_{esc} \simeq \sqrt{6c}\, V_{vir}$$

c is concentration parameter.

So $$E_{ej} \simeq 3c\, M_{ej} V_{vir}^2$$

Kinetic supernova feedback

Kinetic feedback – gas is ejected out of the dark matter halo and can not form future stars.

The energy available from supernovae

$$E_{fb} = \varepsilon_{SN} \xi M_* E_{SN}$$

$\varepsilon_{SN} \lesssim 1$ fraction of total SN energy avaible

$\xi \simeq 0.01 M_\odot^{-1}$ => number of SN produced per Solar mass

$E_{SN} = 10^{51}$ erg energy supplied per SN

Kinetic supernova feedback

Kinetic feedback – gas is ejected out of the dark matter halo and can not form future stars.

Equeting E_{sb} to E_{ej} we can obtain:

$$E_{FB} = E_{ej}$$

$$\varepsilon_{SN} \{ M_* E_{SN} = 3_c M_{ej} V_{vir}^2$$

$$\frac{M_{ej}}{M_*} = 0.4\, \varepsilon_{SN} \left(\frac{c}{10}\right)^{-1} \left(\frac{V_{vir}}{200\ km/s}\right)^{-2}$$

MW ~ $c=10$, $V_{vir} = 200$ km/s

$M_{ej} \sim 0.4\, M_*$ if all SNe energy used

Dwarf galaxy ~ $c \sim 10$; $V_{vir} = 50$ km/s

$M_{ej} \sim 6.4\, M_*$ if all SNe energy used.

Not very efficient in MW!

Thermal supernova feedback

Thermal feedback – Just prevent the gas from cooling to a temperature where it can form stars

Let's assume we want to reheat gas virial T.

Internal energy of gas:

$$E_{int} = \frac{3}{2} M_{gas} \frac{k_B T}{\mu m_p}$$

Want to heat from $T_{init} = 10^4 K$ to

$$T_{vir} = \frac{\mu m_p}{2 k_B} V_{vir}^2$$

Thermal supernova feedback

Thermal feedback – Just prevent the gas from cooling to a temperature where it can form stars

This requires an energy of

$$E_{reheat} = \frac{3}{2} M_{gas} \frac{k_B (T_{vir} - T_{init})}{\mu m_p} = \frac{3}{4} M_{gas} V_{vir}^2 \left[1 - \frac{T_{init}}{T_{vir}} \right]$$

Equate E_{reheat} to E_{FB}

$$\frac{M_{gas}}{M_*} \simeq 17 \, \varepsilon_{SN} \left(\frac{V_{vir}}{200 \text{ km/s}} \right)^{-2} \left(1 - \frac{T_{init}}{T_{vir}} \right)^{-1}$$

1 Solar mass of stars can prevent 17 $M_\odot$ of gas from cooling.

Thermal supernova feedback

Thermal feedback – Just prevent the gas from cooling to a temperature where it can form stars

Reheating is more efficient than ejection by a factor of

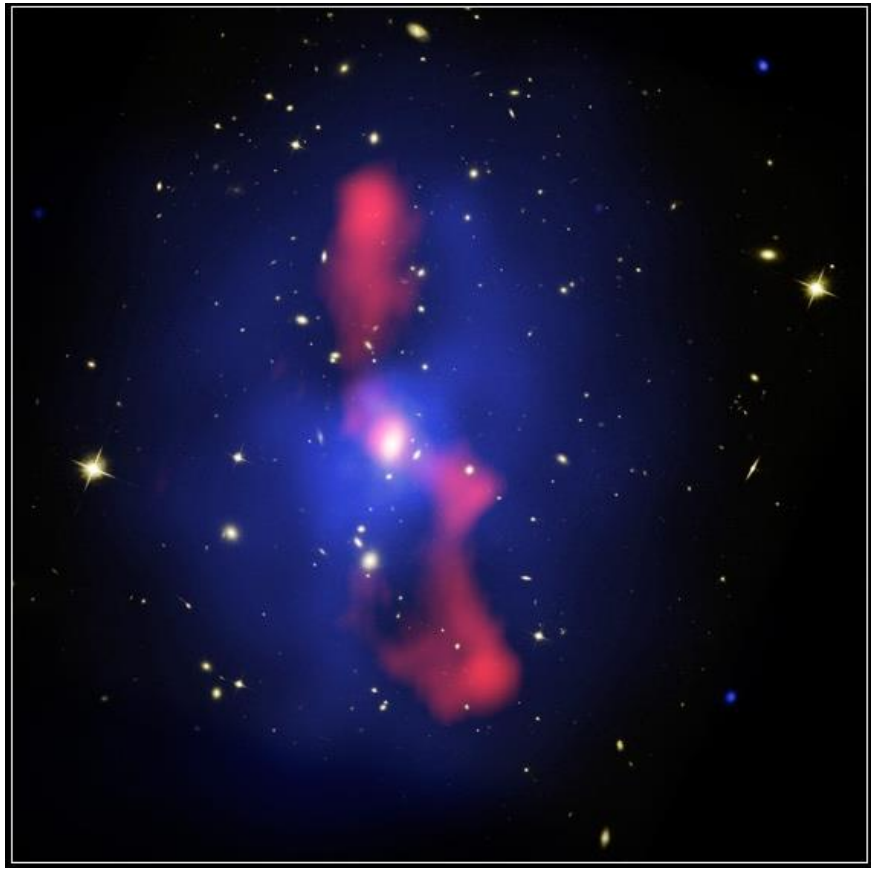
$$\left(\frac{V_{esc}}{V_{vir}}\right)^2 \simeq 6c$$

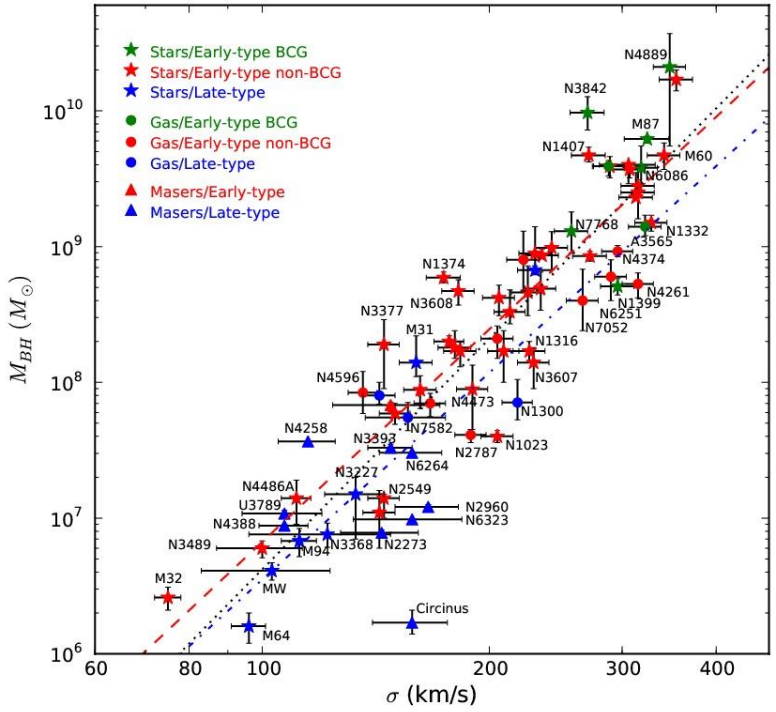
Unsolved problem: what is ε_{SN}

could be $0.01 \leq \varepsilon_{SN} \leq 1$.

Observations of black hole feedback

Again, direct observations are difficult, but there are outflows which suggest a black hole origin, and triggered explosions in simulations can mimic that. The existence of the black hole scaling relations is also strong motivation for an important role.





Black hole/AGN feedback

At high stellar masses, the supernova becomes inefficient to stop star formation.
Black holes potentially have more energy to tap

But, this is likely still not enough. we need AGN feedback.

$$E_{AGN} = 0.1\, M_{BH}\, c^2$$

i.e. 10% of rest mass energy is radiated away.

As we will see $M_{BH} \simeq 0.002\, M_{Bulge}$

Black hole/AGN feedback

At high stellar masses, the supernova becomes inefficient to stop star formation.
Black holes potentially have more energy to tap

Therefore, the feedback from AGN can be related to that of SN by:

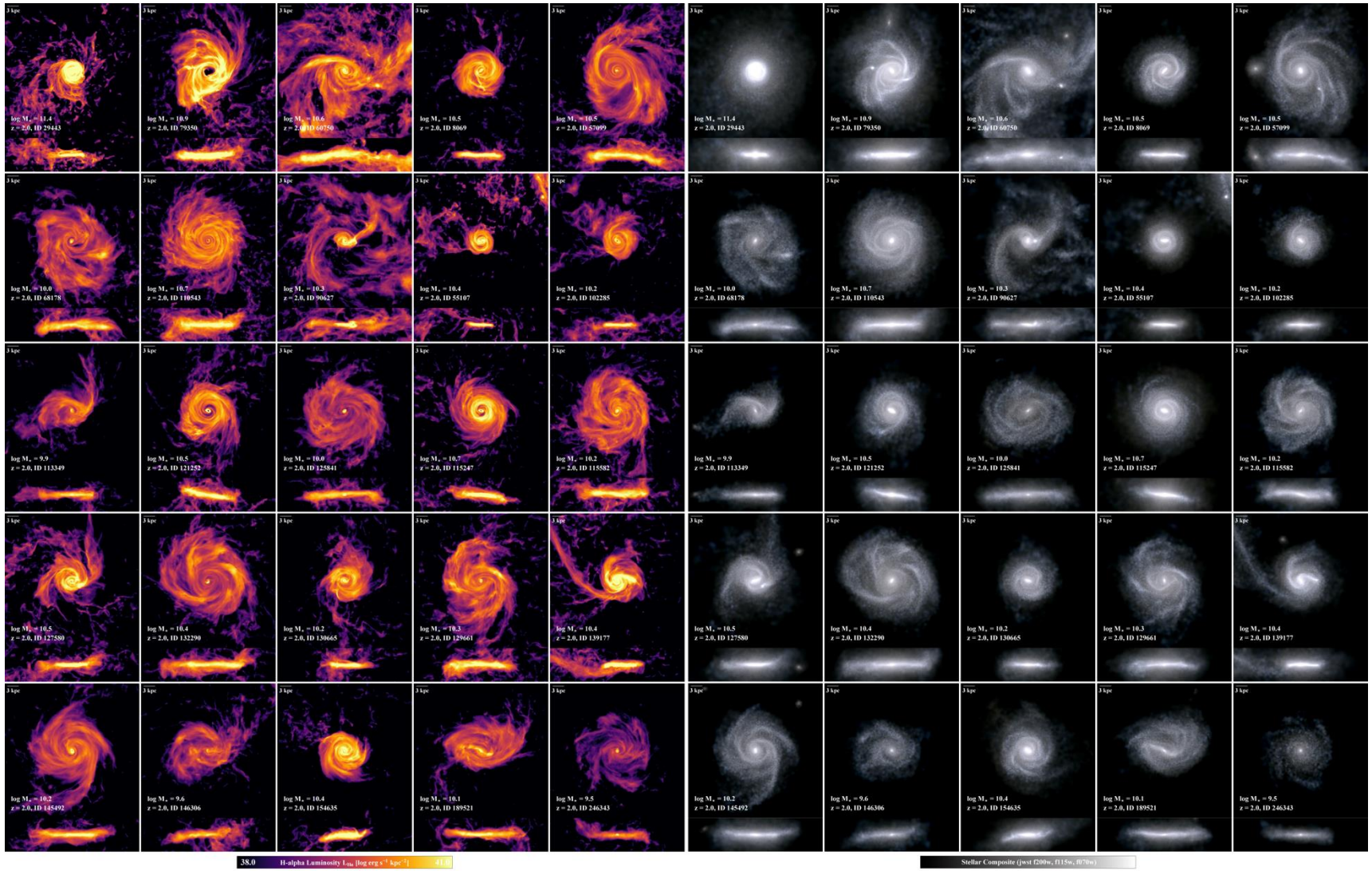
$$\frac{E_{AGN}}{E_{SN}} \sim 36 \frac{\varepsilon_{AGN}}{\varepsilon_{SN}} \left(\frac{M_{bulge}}{M_{\star}} \right)$$

if $\varepsilon_{AGN} \sim \varepsilon_{SN}$ then AGN is order of magnitude more efficient.

Also would help explain scaling relations.

Overview of galaxy formation

That’s the overview of how galaxies form!



Consider a galaxy cluster at $z = 0$ in a flat universe ($\Omega_{M,0} = 0.3$ and $H_0 = 70$ km s^{-1} Mpc^{-1}). This cluster has a baryon fraction f_b of 0.15, a stellar fraction f_* of 0.03 and a total mass of 1×10^{15} M$_\odot$. The gas in the cluster has an observed temperature of 2.5×10^7 K and is of primordial abundance.

(a) If this cluster obeyed the virial theorem, the gas temperature would be 1.0×10^7 K. How much extra energy is contained in this gas?

(b) Assume the stellar content formed with an initial mass function where 0.01 supernovae were produced for every solar mass of stars formed. Calculate the maximum energy that supernovae could deposit into the cluster gas. Recall that a single supernova has an energy of 10^{51} ergs.

(c) If 80% of the stars are contained within galaxy bulges, what is the maximum energy available for AGN feedback in this cluster? Recall that black hole scaling relations find the central black hole in a galaxy is 1% of the mass of the galaxy's bulge.

Consider a galaxy cluster at $z = 0$ in a flat universe ($\Omega_{M,0} = 0.3$ and $H_0 = 70$ km s^{-1} Mpc^{-1}). This cluster has a baryon fraction f_b of 0.15, a stellar fraction f_* of 0.03 and a total mass of 1×10^{15} M$_\odot$. The gas in the cluster has an observed temperature of 2.5×10^7 K and is of primordial abundance.

(a) If this cluster obeyed the virial theorem, the gas temperature would be 1.0×10^7 K. How much extra energy is contained in this gas?

Answer: The internal energy of the gas is $E_{int} = \frac{3}{2} M_g \frac{k_b T}{\mu m_p}$. So the extra energy is

$$E = \frac{3}{2} M_g \frac{k_b (T_f - T_i)}{\mu m_p} \tag{6}$$

The gas mass can be found

$$M_g = (f_b - f_*) M_{\text{cluster}} = 1.2 \times 10^{14} M_\odot \tag{7}$$

Assuming a Helium of 25% and hydrogen of 75% this energy can be calculated and found to be 1.8×10^{63} ergs.

Consider a galaxy cluster at $z = 0$ in a flat universe ($\Omega_{M,0} = 0.3$ and $H_0 = 70$ km s^{-1} Mpc^{-1}). This cluster has a baryon fraction f_b of 0.15, a stellar fraction f_* of 0.03 and a total mass of 1×10^{15} M$_\odot$. The gas in the cluster has an observed temperature of 2.5×10^7 K and is of primordial abundance.

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Answer: The supernova energy from a population of stars can be found as

$$E = \eta M_* E_{sn} \tag{8}$$

where M_* is the total mass in stars, η is the number of supernova produced per solar mass of star formed and E_{sn} is the energy of a single supernova. To find M_* we calculate

$$M_* = f_* M_{\text{cluster}} = 3 \times 10^{13} M_\odot \tag{9}$$

and therefore the total energy is

$$E = \eta M_* E_{sn} = 3 \times 10^{62} ergs \tag{10}$$

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(c) If 80% of the stars are contained within galaxy bulges, what is the maximum energy available for AGN feedback in this cluster? Recall that black hole scaling relations find the central black hole in a galaxy is 1% of the mass of the galaxy's bulge.

Answer: The energy available from a black hole is

$$E = M_{bh}c^2 \tag{11}$$

The mass in black holes can be found as

$$M_{bh} = 0.01 * 0.8 * f_* M_{\text{cluster}} = 2.4 \times 10^{11} M_\odot \tag{12}$$

So, the total energy is $M_{bh}c^2 = 4.0 \times 10^{65}$ ergs.